Documentation for Search Algorithms

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Uninformed Search Algorithms Documentation

This document explains four uninformed search algorithms implemented for the 8-5-3 Milk Jug Problem: Breadth-First Search (BFS), Depth-First Search (DFS), Depth-Limited Search (DLS), and Iterative Deepening Depth-First Search (IDDFS).

Problem:

The 8–5–3 Milk Jug Problem is a problem used in Artificial Intelligence to demonstrate search algorithms. In this problem, you are given three jugs with capacities of 8 liters, 5 liters, and 3 liters. Initially, the 8-liter jug is full of milk, and the other two jugs are empty. The task is to measure exactly 4 liters in the 5-liter jug by pouring milk between the jugs. You are only allowed to pour milk from one jug to another until either the source jug becomes empty or the destination jug becomes full, and you cannot spill any milk. The challenge is to find a sequence of valid pouring steps that achieves this goal.

1. Breadth-First Search (BFS)

How It Works:

BFS explores the state space level by level. It first explores all nodes at depth d before moving to nodes at depth d+1. It guarantees the shortest path in terms of number of steps.

Data Structure Used:

• Queue (FIFO - First In First Out)

Properties:

• Optimal but uses a lot of memory

• Time Complexity: O(b^d)

• Space Complexity: O(b^d)

where b = branching factor, d = depth of solution

2. Depth-First Search (DFS)

How It Works:

DFS explores as deep as possible along one branch before backtracking. It does not guarantee the shortest solution.

Data Structure Used:

• Stack (LIFO - Last In First Out) or Recursion

Properties:

• No Completeness (if infinite depth)

• Low optimality, but low amount of memory also used.

• Time Complexity: O(b^m)

• Space Complexity: O(bm)

where m = maximum depth of the tree

3. Depth-Limited Search (DLS)

How It Works:

DLS is a modified DFS that introduces a depth limit. It prevents infinite recursion by restricting the maximum search depth.

Data Structure Used:

• Stack / Recursion with depth parameter

Properties:

• Optimality better than DFS but still lacking

• Time Complexity: O(b^l)

• Space Complexity: O(bl)

where l = depth limit

4. Iterative Deepening DFS (IDDFS)

How It Works:

IDDFS repeatedly applies Depth-Limited Search with increasing depth limits starting from 0 until the goal is found. It combines the advantages of BFS and DFS by being optimal and memory efficient.

Data Structure Used:

• Stack / Recursion (Repeated Depth-Limited Search)

Properties:

- Very optimal
- Time Complexity: O(b^d)
- Space Complexity: O(bd)

5. How to run the codes:

The codes are in the folder, and each have to be executed respectively using Python3.

Conclusion

For the 8-5-3 Milk Jug Problem, BFS guarantees the shortest solution but uses more memory. DFS uses less memory but is not optimal. DLS prevents infinite depth issues but depends on the chosen depth limit. IDDFS provides the best trade-off by being both complete and optimal while using significantly less memory than BFS.